

Department of Computer Science and Engineering

Assignment Questions

Course Code & Name	ITA0608 – Machine Learning
Name/Reg	Thamizharasan S (192424087)
Assignment Title	Intelligent Crop Yield Prediction and Agricultural Risk Classification System Using Machine Learning
Course Outcome(s)	CO1: Version spaces, candidate elimination, inductive bias, heuristic search. CO2: Learning paradigms, decision tree learning, concept representation. CO3: Neural network architectures — perceptron, MLP, backpropagation. CO4: Genetic algorithms/programming for hypothesis search and model evaluation. CO5: Bayesian methods — Bayes theorem, MLE, Bayesian classifiers.
Bloom's Taxonomy Level	L3 – Apply; L4 – Analyse; L5 – Evaluate; L6 – Create
SDG Mapping	SDG 2 – Zero Hunger SDG 13 – Climate Action SDG 15 – Life on Land

Problem Statement

Design, implement, and evaluate an Intelligent Crop Yield Prediction and Agricultural Risk Classification System using machine learning techniques.

Agricultural organizations and farm advisory bodies need tools that assist agronomists and farmers in analysing farm-level data and predicting potential crop-yield risk. The system uses an agricultural dataset containing attributes such as rainfall, temperature, humidity, soil pH, nitrogen (N), phosphorus (P), potassium (K) levels, farm area, fertilizer usage, irrigation coverage, pesticide usage, and cropping season.

The proposed system analyses the farm data and classifies each record into an appropriate agricultural risk category — Low Risk, Moderate Risk, or High Risk — indicating the likelihood of a poor crop yield outcome for the given season.

The final system integrates the following machine learning workflow:

Agricultural Dataset → Data Preprocessing → Statistical Analysis → Feature Selection → Model Development → Prediction → Model Evaluation → Risk Classification

Students should compare the performance of the implemented approaches and identify the most suitable technique for the selected agricultural prediction problem. The system should provide meaningful outputs such as:

- Predicted crop-risk category (Low / Moderate / High)
- Important agro-climatic and soil attributes
- Model accuracy and other evaluation measures
- Comparison of different machine learning algorithms
- Visual representation of prediction results

Farms generate rich seasonal data — rainfall, temperature, soil nutrients, irrigation coverage, and input usage. Manual assessment of yield risk across many farms is slow and inconsistent.

Machine learning can assist by discovering patterns in this agro-climatic data and predicting risk categories automatically. This project develops an Intelligent Crop Yield Prediction and Agricultural Risk Classification System using Machine Learning.

The collected data is preprocessed to handle missing values and inconsistencies, then analysed using NumPy and Pandas to study distributions and relationships among agro-climatic attributes. Statistical, correlation, and visualization analysis is used to uncover useful patterns, followed by feature selection to retain the most relevant attributes.

Multiple machine learning techniques are used to build and compare prediction models: a Decision Tree for interpretable decision rules, a Multilayer Perceptron (MLP) trained with backpropagation for capturing nonlinear relationships, and a Gaussian Naive Bayes classifier for probabilistic prediction. A Genetic Algorithm is additionally applied for feature-subset optimization.

Farms are classified into Low Risk, Moderate Risk, or High Risk categories. Models are evaluated and compared using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC, and visualizations are produced for feature importance, model performance, and model comparison. This project supports SDG 2 – Zero Hunger, SDG 13 – Climate Action, and SDG 15 – Life on Land.

Pseudocode

Algorithm: Intelligent Crop Yield Prediction and Agricultural Risk Classification

Input: Agricultural (crop-yield-risk) dataset with rainfall, temperature, humidity, soil and nutrient attributes

Output: Predicted agricultural risk category and model performance comparison

1. Start
2. Load the agricultural crop-yield dataset.
3. Separate the dataset into feature variables X and target variable Y (risk category).
4. Define the target as three classes: 0 = Low Risk, 1 = Moderate Risk, 2 = High Risk.
5. Perform data preprocessing: convert feature values into numeric format; handle missing values using median imputation; split the dataset into training and testing sets; standardize the feature values.
6. Perform statistical analysis: calculate descriptive statistics; analyze the risk-class distribution; generate a correlation matrix.
7. Perform feature selection using Mutual Information; calculate the importance score of each feature; select the top relevant features.
8. Train a Decision Tree classifier using the selected features.
9. Train an MLP Neural Network using the selected features.
10. Train a Gaussian Naive Bayes classifier using the selected features.
11. Apply a Genetic Algorithm for feature selection: create an initial population of feature combinations; evaluate each combination using classification accuracy; select the best individuals; apply crossover and mutation; repeat for multiple generations; select the best feature subset.

12. Train an MLP classifier using the GA-selected features.
13. Evaluate all models using accuracy, precision, recall, F1-score, confusion matrix, ROC curve and AUC.
14. Compare the performance of all models.
15. Identify the best-performing model based on accuracy and other evaluation metrics.
16. Provide a sample farm record to the selected GA-optimized model.
17. Predict the farm's risk category: Low Risk, Moderate Risk, or High Risk.
18. Display the prediction confidence.
19. Save the model performance results to a CSV file.
20. Stop

Dataset

The project uses an Agricultural Crop-Yield-Risk dataset representing seasonal farm records across multiple regions. It consists of 12 agro-climatic and soil features, including rainfall, temperature, humidity, soil pH, nitrogen, phosphorus, potassium, farm area, fertilizer use, irrigation percentage, pesticide use, and cropping season.

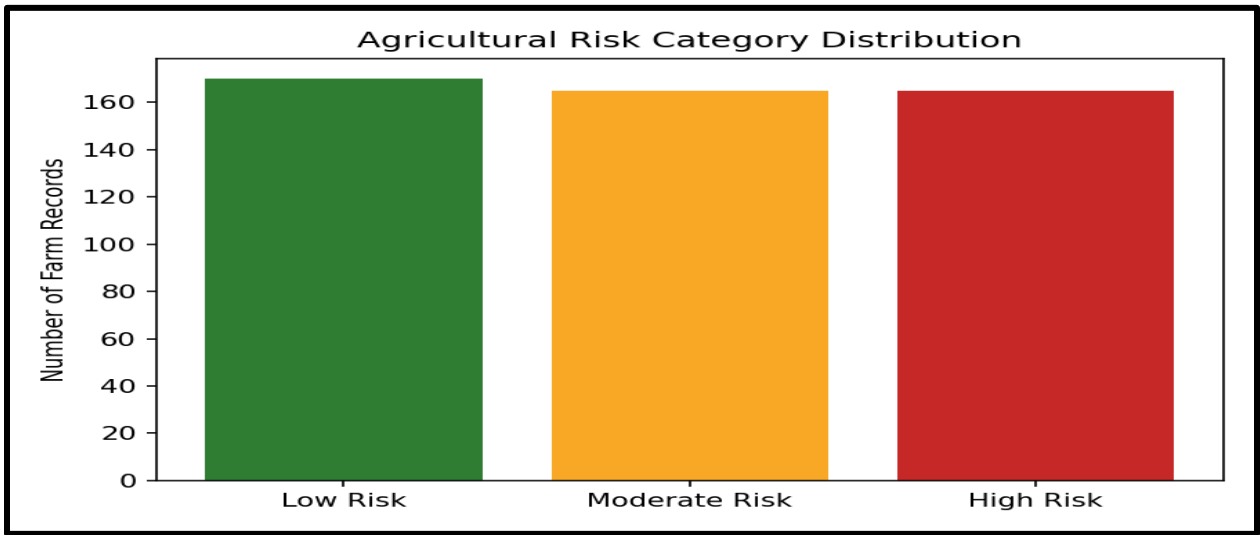
The target variable is a three-class agricultural risk category, where the underlying seasonal yield outcome is derived from agro-climatic conditions and input usage, and converted into Low Risk, Moderate Risk, and High Risk classes using tertile thresholds. Missing values are handled using median imputation, and the input features are standardized before model training.

The processed dataset is divided into training and testing subsets. The training data is used to develop the Decision Tree, MLP Neural Network, Bayesian Classifier, and Genetic Algorithm-optimized MLP models. The testing data is used to evaluate and compare their performance using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC.

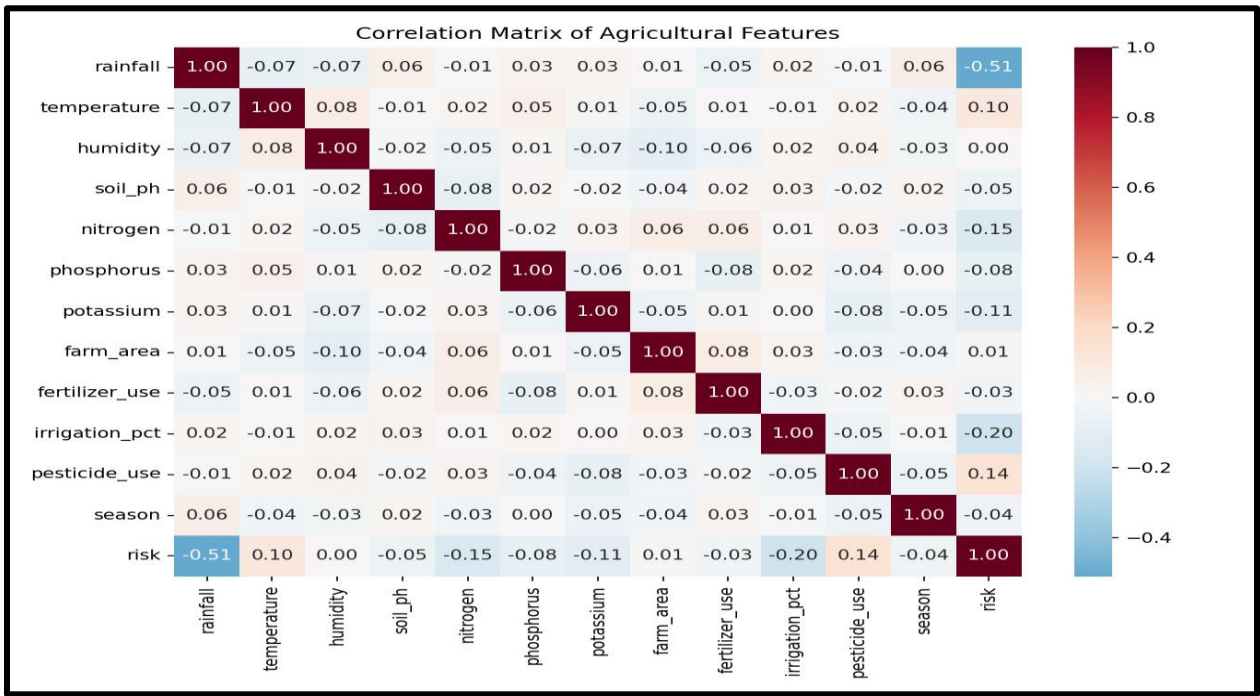
Dataset Name	Agricultural Crop-Yield-Risk Dataset (synthetic, agro-climatic)
Problem Type	Multi-class Classification (3 classes)
Number of Input Features	12
Target Classes	Low Risk / Moderate Risk / High Risk
Records	500 (400 training / 100 testing)

Results and Discussion

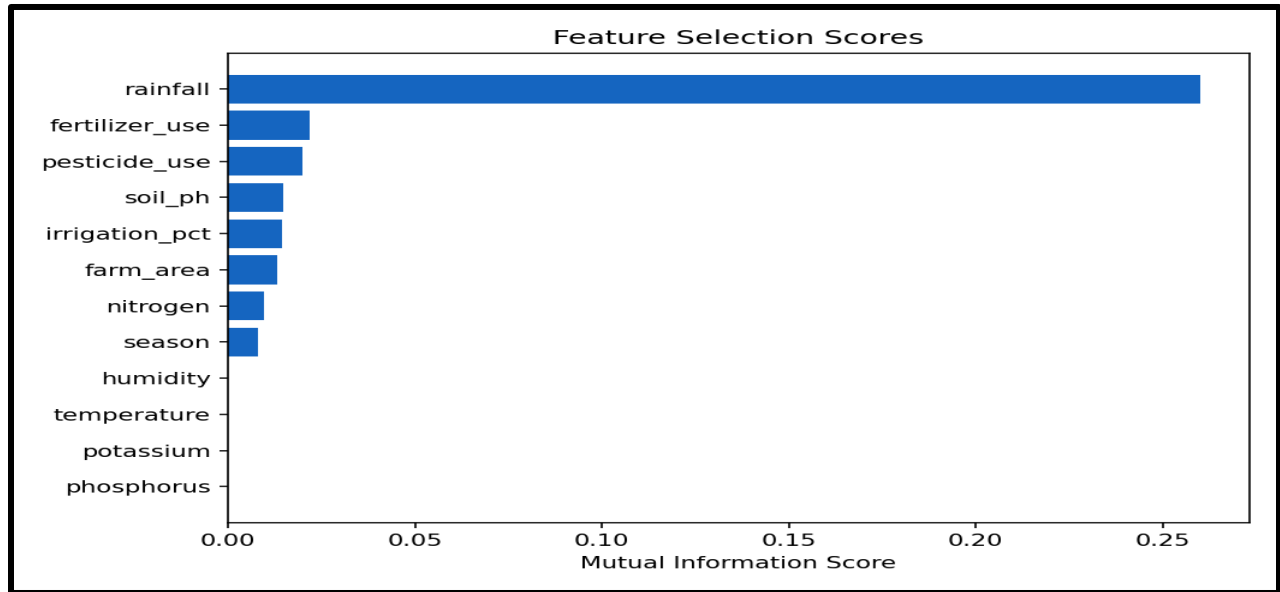
Risk class distribution across the dataset:



Correlation matrix of agricultural features:



Feature selection scores (Mutual Information). Rainfall was found to carry by far the strongest association with yield risk, followed by fertilizer use, pesticide use, soil pH, irrigation coverage, farm area, and nitrogen:

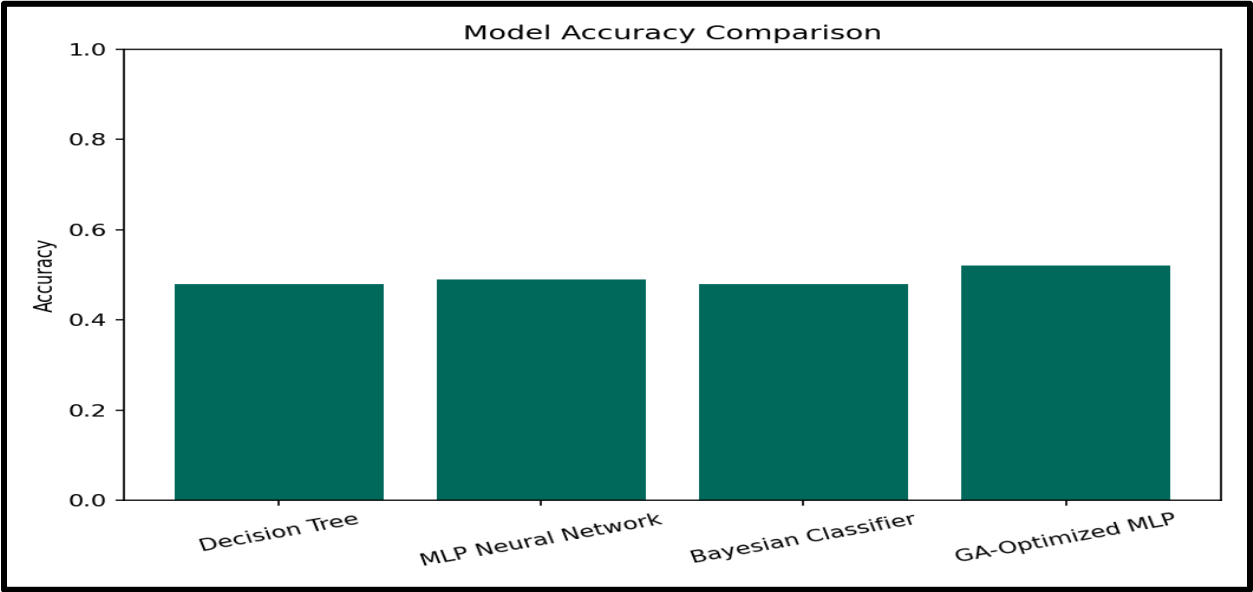


The Genetic Algorithm converged on a compact 3-feature subset — rainfall, temperature, and pesticide use — for the GA-optimized MLP.

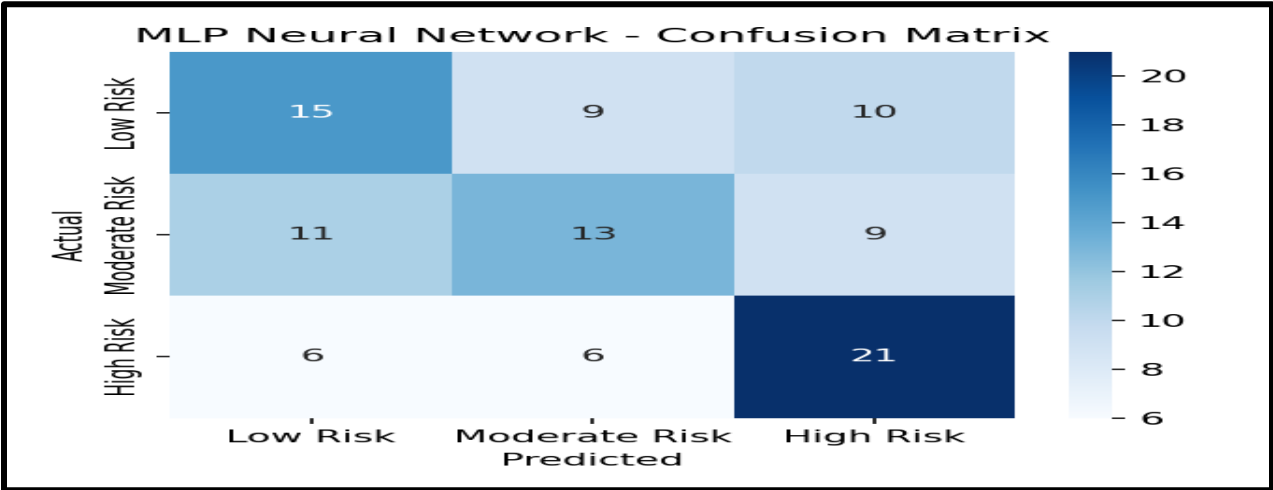
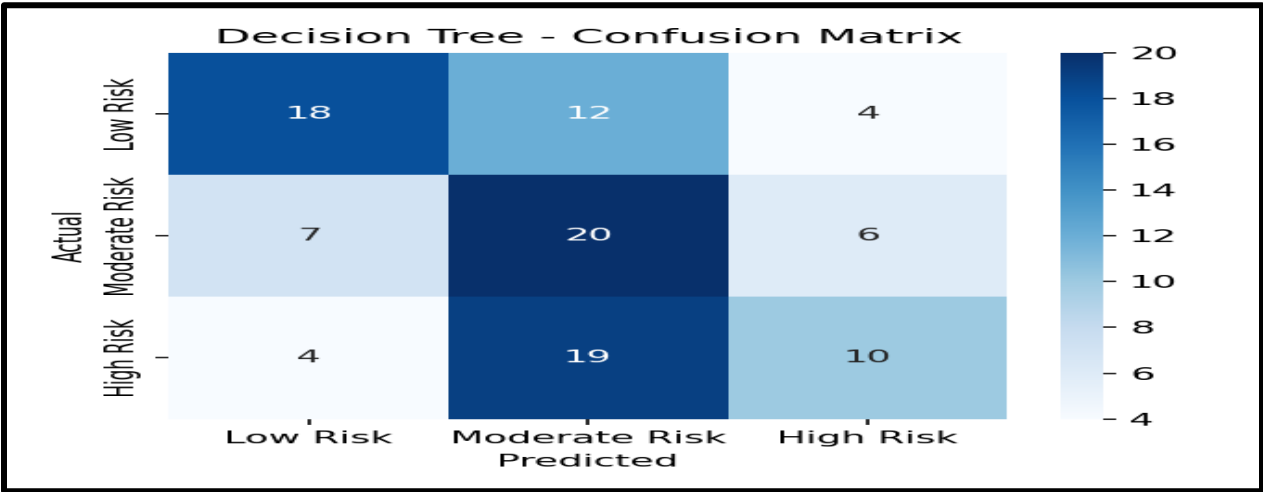
Model performance on the held-out test set:

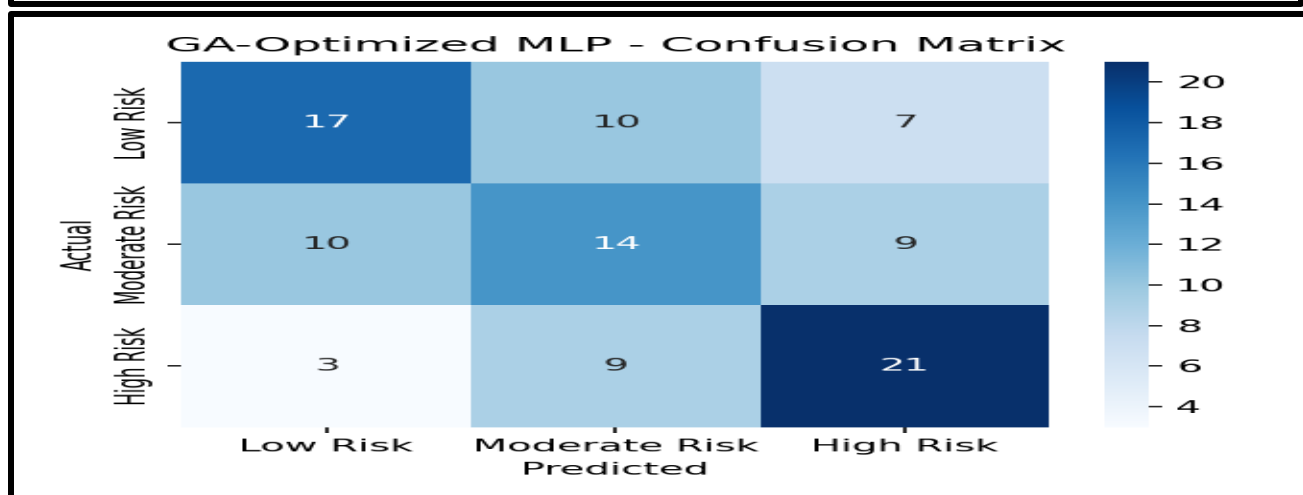
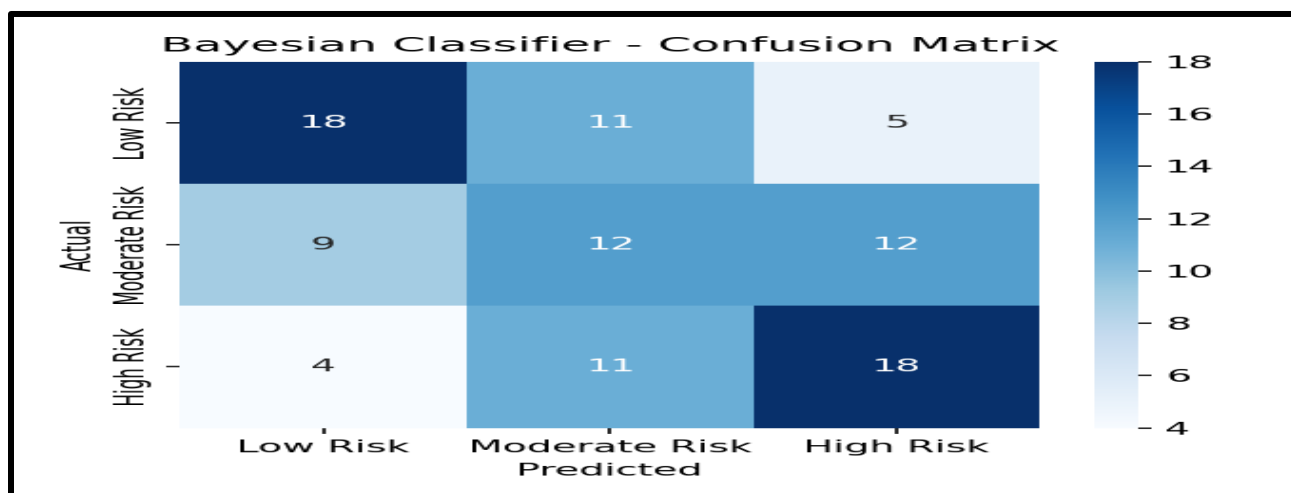
Model	Accuracy	Precision	Recall	F1-Score
Decision Tree	0.4800	0.5043	0.4795	0.4750
MLP Neural Network	0.4900	0.4860	0.4905	0.4854
Bayesian Classifier	0.4800	0.4826	0.4795	0.4805
GA-Optimized MLP	0.5200	0.5195	0.5202	0.5185

Model accuracy comparison:

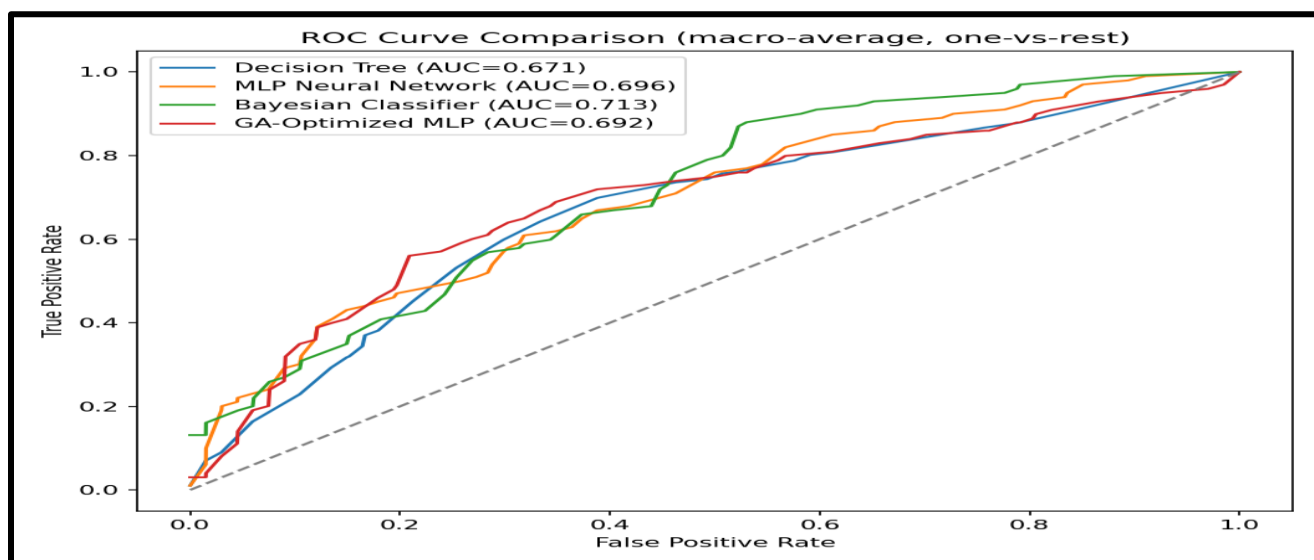


Confusion matrices for each classifier:



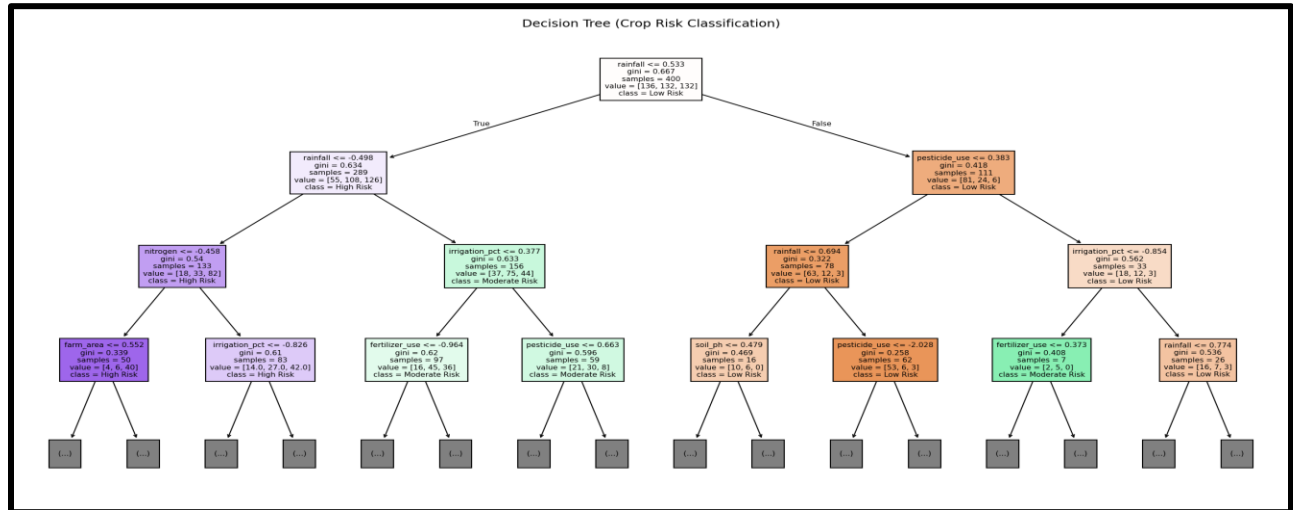


ROC curve comparison (macro-averaged, one-vs-rest, across the three risk classes):

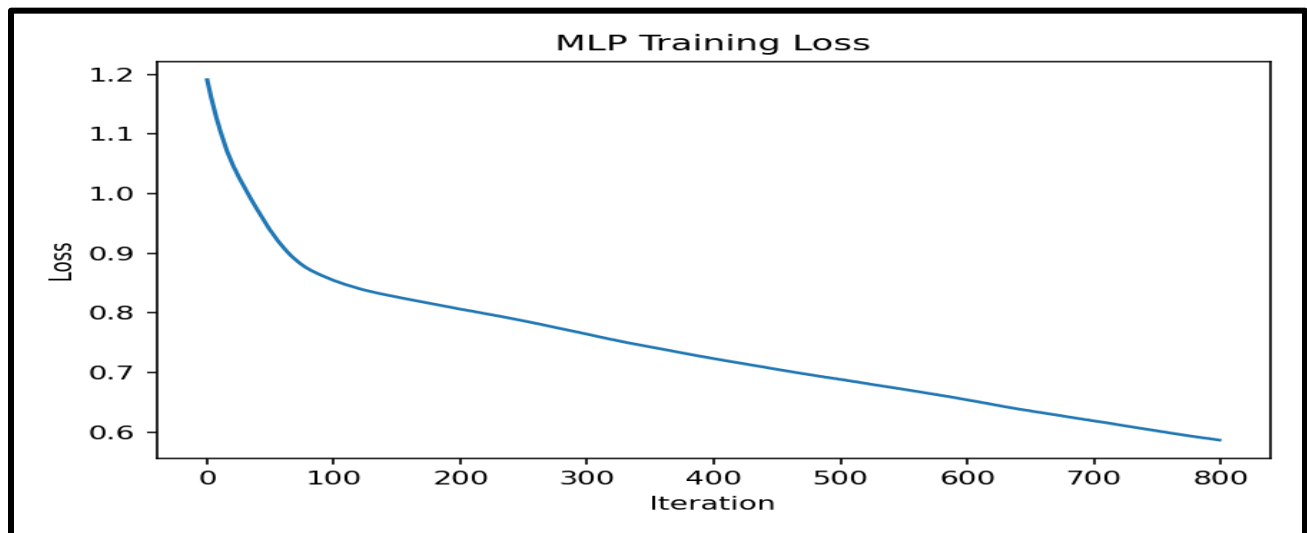


AUC values obtained: Decision Tree = 0.671, MLP Neural Network = 0.696, Bayesian Classifier = 0.713, GA-Optimized MLP = 0.692.

Decision tree structure (top levels) learned on the Mutual-Information-selected features:



MLP training loss curve:



Sample farm risk prediction using the GA-optimized MLP — a farm with low rainfall (620 mm), high temperature (34°C), low irrigation coverage (15%) and low soil nutrients was submitted to the trained model:

Prediction: High Risk | Confidence: 52.59%

Methodology

The proposed system follows a machine learning-based agricultural risk prediction workflow for classifying farm-season records into crop-yield risk categories. The methodology consists of data preprocessing, statistical analysis, feature selection, classification, Genetic Algorithm-based optimization, and performance evaluation.

1. Data Preprocessing

The agricultural dataset is loaded and separated into input features and the target variable. The target is defined as three risk classes derived from the underlying agro-climatic yield outcome. Missing numerical values (in rainfall, soil pH, and nitrogen) are handled using median imputation. The dataset is split into training and testing sets, and feature standardization is applied.

2. Statistical and Exploratory Analysis

Descriptive statistics are calculated to understand the distribution of agro-climatic attributes. The distribution of Low, Moderate, and High risk cases is visualized to examine class balance, and a correlation matrix is generated to identify relationships among rainfall, soil nutrients, irrigation, and other inputs.

3. Feature Selection

Mutual Information is applied to measure the relevance of individual agro-climatic features with respect to the risk category. The most informative features — rainfall, fertilizer use, pesticide use, soil pH, irrigation percentage, farm area, and nitrogen — are retained for model development.

4. Classification Models

Decision Tree: divides farm records according to feature-value conditions and produces a risk classification based on the resulting decision paths.

MLP Neural Network: learns nonlinear relationships between the selected agro-climatic features and the risk classes through hidden layers trained with backpropagation.

Bayesian Classifier: a Gaussian Naive Bayes model estimates the probability of each risk class from the observed agricultural attributes.

5. Genetic Algorithm Optimization

A Genetic Algorithm searches for an optimized subset of agro-climatic features. Each feature combination is represented as a binary individual; fitness is computed from classification accuracy with a small penalty for excessive features. Tournament-style survivor selection,

single-point crossover, and bit-flip mutation are applied over multiple generations, converging on rainfall, temperature, and pesticide use as the optimized subset.

6. GA-Optimized MLP

The GA-selected features are used to train a second MLP, which is evaluated against the Decision Tree, standard MLP, and Bayesian classifier to determine whether GA-based feature optimization improves prediction performance.

7. Performance Evaluation

All models are evaluated using accuracy, precision, recall, and F1-score (macro-averaged across the three risk classes). Confusion matrices and macro-averaged ROC-AUC (one-vs-rest) are used to further assess classification quality.

8. Farm Risk Prediction

A sample farm record is passed to the GA-optimized MLP, which predicts the risk category and reports a prediction confidence, demonstrating the practical use of the system for agricultural decision support.

Results and Performance Evaluation

The developed system was evaluated on a held-out test set of 100 farm-season records. Four classification approaches were considered: Decision Tree, MLP Neural Network, Bayesian Classifier, and Genetic Algorithm-optimized MLP.

The GA-Optimized MLP achieved the highest accuracy of 52.00%, with precision 0.5195, recall 0.5202, and F1-score 0.5185 (macro-averaged across the three risk classes), outperforming the Decision Tree (48.00%), Bayesian Classifier (48.00%), and standard MLP (49.00%). This indicates that Genetic-Algorithm-based feature optimization provided a modest but consistent improvement over the other approaches on this three-class problem.

Under ROC-AUC analysis, the Bayesian Classifier obtained the highest macro-averaged AUC (0.713), followed by the MLP Neural Network (0.696), GA-Optimized MLP (0.692), and Decision Tree (0.671), suggesting that while the GA-Optimized MLP produced the best hard-label accuracy, the Bayesian Classifier's probability estimates were the most consistently well-ranked across all three risk classes.

Because the classification is a three-class (rather than binary) agricultural risk problem, overall accuracies in the 48–52% range are notably better than the 33% expected from random guessing, but also reflect the genuine difficulty of separating three risk tiers from noisy, overlapping agro-climatic conditions — consistent with real-world crop-risk data. The final model selection is based on the actual evaluation metrics obtained rather than assuming any one algorithm will always perform best.

Conclusion

The Intelligent Crop Yield Prediction and Agricultural Risk Classification system was successfully developed using machine learning and Genetic Algorithm-based feature optimization. The agricultural dataset was preprocessed by handling missing values, standardizing agro-climatic attributes, and selecting relevant features using Mutual Information. Decision Tree, MLP Neural Network, and Bayesian classification techniques were implemented and evaluated for crop-yield risk prediction.

A Genetic Algorithm was applied to identify an optimized subset of agro-climatic features (rainfall, temperature, pesticide use), and the GA-Optimized MLP built on this subset achieved the best overall accuracy among the four models. Performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC analysis.

The system demonstrates that machine learning can be applied to agro-climatic and soil data for automated crop-yield risk classification, and can serve as a decision-support tool for preliminary agricultural risk assessment. However, predictions should not replace agronomic expertise or on-ground assessment. Future improvements can include real farm-sensor and satellite-derived data, additional soil and weather attributes, explainable AI techniques, and deployment as a real-time farm-advisory application.

Assessment Rubrics

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Concept Learning: Version Space, Candidate Elimination, Inductive Bias & Heuristic Search (CO1)	15	Correctly formulates an agricultural concept-learning problem and clearly demonstrates version space, candidate elimination, inductive bias, and heuristic search.	Concepts correctly applied with minor gaps.	Basic concepts demonstrated but limited analysis.	Concepts missing, incorrectly applied, or poorly explained.
Learning Paradigms & Decision Tree Learning (CO2)	15	Correctly implements decision tree learning, clearly represents crop-risk concepts, explains attribute selection and decision rules, and interprets results.	Decision tree correctly implemented with minor analytical gaps.	Basic decision tree implementation with limited interpretation.	Decision tree incomplete, incorrect, or poorly interpreted.

Neural Networks: Perceptron, MLP & Backpropagation (CO3)	15	Successfully implements neural network architecture and backpropagation; clearly analyses training parameters and prediction performance.	Neural network correctly implemented with minor issues.	Basic implementation works but analysis is limited.	Neural network or backpropagation implementation incomplete or incorrect.
Genetic Algorithm / Genetic Programming (CO4)	15	Effectively applies genetic algorithms for hypothesis search, feature selection, or model optimization and evaluates its impact.	Genetic algorithm correctly applied with reasonable analysis.	Basic implementation provided but optimization analysis limited.	Genetic algorithm missing or incorrectly implemented.
Bayesian Learning & Computational Techniques (CO5)	15	Correctly applies Bayes theorem, MLE, and Bayesian classification with meaningful probability-based interpretation.	Bayesian techniques correctly implemented with minor gaps.	Basic Bayesian classification implemented with limited interpretation.	Bayesian methodology missing or incorrectly applied.
Data Manipulation & Analysis Using NumPy and Pandas (CO7)	15	Efficiently performs preprocessing, statistical analysis, aggregation, feature preparation, correlation analysis, and visualization.	Most data-analysis tasks correctly completed with minor issues.	Basic preprocessing and analysis completed but lack depth.	Data manipulation and analysis incomplete or contain significant errors.
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Thoughtful justification of design choices, clear connection to SDG 2/13/15 relevance, and honest, specific account of challenges and learning.	General justification; sustainability connection and challenges mentioned but lack depth.	Reflection present but generic; limited justification or vague account.	No genuine reflection, or generic content with no real connection.

Deliverables

To receive full credit, each submission must include the following deliverables:

- 1. Problem Definition
- 2. Pseudocode
- 3. Results and Discussion
- 4. Final Report
- 5. GitHub Upload